For the partition $\lambda = (1, \dots, 1)$, $Q_\lambda = S_n$, $P_\lambda = \{1\}$, so c_λ is the antisymmetrizer, and hence V_λ is the sign representation.

$n = 3$. For $\lambda = (2, 1)$, $V_\lambda = \mathbb{C}^2$.

$n = 4$. For $\lambda = (2, 2)$, $V_\lambda = \mathbb{C}^2$; for $\lambda = (3, 1)$, $V_\lambda = \mathbb{C}^3_-$; for $\lambda = (2, 1, 1)$, $V_\lambda = \mathbb{C}^3_+$.

Corollary 5.12.4. *All irreducible representations of S_n can be given by matrices with rational entries.*

Problem 5.12.5. Find the sum of dimensions of all irreducible representations of the symmetric group S_n.

Hint: Show that all irreducible representations of S_n are real, i.e., admit a nondegenerate invariant symmetric form. Then use the Frobenius-Schur theorem.

5.13. Proof of the classification theorem for representations of S_n

Lemma 5.13.1. *Let $x \in \mathbb{C}[S_n]$. Then $a_\lambda x b_\lambda = \ell_\lambda(x) c_\lambda$, where ℓ_λ is a linear function.*

Proof. If $g \in P_\lambda Q_\lambda$, then g has a unique representation as pq, $p \in P_\lambda, q \in Q_\lambda$, so $a_\lambda g b_\lambda = (-1)^q c_\lambda$. Thus, to prove the required statement, we need to show that if g is a permutation which is not in $P_\lambda Q_\lambda$, then $a_\lambda g b_\lambda = 0$.

To show this, it is sufficient to find a transposition t such that $t \in P_\lambda$ and $g^{-1} t g \in Q_\lambda$; then

$$a_\lambda g b_\lambda = a_\lambda t g b_\lambda = a_\lambda g (g^{-1} t g) b_\lambda = -a_\lambda g b_\lambda,$$

so $a_\lambda g b_\lambda = 0$. In other words, we have to find two elements i, j standing in the same row in the tableau $T = T_\lambda$ and in the same column in the tableau $T' = gT$ (where gT is the tableau of the same shape as T obtained by permuting the entries of T by the permutation g). Thus, it suffices to show that if such a pair does not exist, then $g \in P_\lambda Q_\lambda$, i.e., there exists $p \in P_\lambda$, $q' \in Q'_\lambda := g Q_\lambda g^{-1}$ such that $pT = q'T'$ (so that $g = pq^{-1}, q = g^{-1} q' g \in Q_\lambda$).